

ADETUNJI SAMUEL

SOFTWARE ENGINEER (AI Agents & Infrastructure)

Contact: samueladetunji000@gmail.com | [Github](#) | +234 8137440918

I am a Software Engineer specializing in Python, TypeScript, and AI infrastructure. I am deeply focused on building durable agentic workflows, orchestrating LLMs in production, and managing complex concurrency in distributed systems. Proven track record of turning complex asynchronous problems such as multi-step agent execution, stateful memory management, and connection pool starvation into clean, scalable developer abstractions.

PROFESSIONAL WORK EXPERIENCE

BuildonInc - Delaware US Senior Full-Stack AI Engineer

April 2026 - till present

- Developed and integrated robust AI agent workflows into enterprise platforms, heavily utilizing TypeScript and Python to manage complex tool-use, stateful memory, and context windows.
- Designed clean, developer-friendly internal APIs over complex backend microservices, enabling rapid iteration and seamless agent-to-system interactions.
- Manage robust CI/CD pipelines, Docker containerization, and cloud infrastructure, ensuring zero-downtime upgrades and resilient execution for asynchronous data-heavy workloads.

Technophlix - India Software Engineer

July 2024 - October 2024

- Collaborated on secure RESTful API development (Node.js/Express) and relational database integration, maintaining seamless, low-latency data flow for production-level traffic.
- Supported CI/CD pipelines via GitHub Actions and deployed backend applications on AWS (EC2/S3), maintaining 99.9% uptime.

Independent Freelancer (Software Engineer)

Jan 2021 - April 2026

- Architected the backend platform for full-stack SaaS applications (including the July 2026 release of DBChores), taking full ownership of authentication, API architecture, and database design.
- Engineered the complete subscription billing lifecycle for a high-traffic URL Shortener, designing idempotent Node.js webhook listeners to reconcile external payment states with internal PostgreSQL databases, ensuring users were never double-charged during latency spikes.
- Built and integrated robust B2B endpoints, ensuring external integrations were heavily monitored and scalable.

PERSONAL PROJECTS

PAAS(BYOC model)

- Architected a complex BYOC infrastructure platform enabling users to orchestrate and provision custom virtual private servers.
- Engineered a high-performance Python (FastAPI) and Node.js backend to handle asynchronous server provisioning, resolving concurrent SSL socket collisions through isolated dependency injection.
- Implemented custom user interfaces in React/TypeScript to consume real-time WebSocket telemetry, rendering live server metrics without degrading frontend performance.

Agentic Workflow Orchestrator Micro-SaaS

- Architected a production-grade multi-tenant AI workflow platform using Python (FastAPI, PydanticAI) and TypeScript (Next.js) to orchestrate complex, multi-step LLM tasks.
- Engineered durable agentic workflows capable of maintaining state and coordinating work across extended interactions, implementing a custom event-driven state machine to prevent context drift.
- Built robust retrieval layers and structured tool-calling pipelines, integrating dynamically with foundational models (OpenAI, Groq) via clean internal APIs.

TECH STACKS(MY SKILL SETS)

- **Backend & Orchestration:** Go (Golang), Node.js (Express), Python (FastAPI), RESTful API Design, Event Processing, WebSockets
- **Frontend & Experience:** TypeScript, JavaScript (ES6+), React.js, Next.js, State Management (Context API, Zustand), UI/UX implementation, Tanstack Query, Global State Management
- **Agentic Infrastructure:** Stateful Workflows, Context/Memory Management, Tool-Calling, RAG, PydanticAI
- **Applied AI:** Agentic Workflows, RAG, Tool-Calling, Prompting, LangChain concepts, LLM APIs (OpenAI, Groq)
- **Databases & Infrastructure:** PostgreSQL, MySQL, MongoDB, Supabase, Prisma, Relational Database Design, Docker, Kubernetes concepts, AWS, GCP, CI/CD, Redis
- **Testing & DevOps:** CI/CD Pipelines, Git, GitHub Actions, Docker, Cypress, Jest, AWS (EC2/S3)

CERTIFICATIONS & EDUCATION

- **HackerRank Software Engineer Certificate (2026)**
- **Obafemi Awolowo University** | 2020 - 2026(August 30) Bachelor of Medical Rehabilitation